

# Forest Sensor Sprint

Find the smartest place to install a future campus tree sensor.

**Win condition:** Best evidence-based sensor recommendation with accurate measurements and efficient route planning.

## THE SCIENCE YOU ARE USING

**Microclimate varies in meters.** Temperature, humidity, light, and soil moisture can change sharply over short distances. A spot under a dense canopy can be several degrees cooler and far more humid than open lawn a few steps away. Good sensor placement starts with measuring, not guessing.

**Evidence-based siting.** A useful sensor sits where its readings represent the area you care about and where it will not be disturbed. Teams collect data at field stations, then argue for one location using their own numbers plus an efficient route between checkpoints.

## HOW TO BUILD AND RUN IT

- 01 Calibrate together.** Take one reading as a whole group at a reference spot so every team's tools agree before they split up.
- 02 Run the field stations.** Rotate through stations measuring temperature, humidity, light, and soil moisture. Record units every time.
- 03 Estimate tree height.** Sight the treetop through the clinometer straw, read the angle off the plumb string, and use the standoff distance the instructor pre-marked with the long tape:  $\text{tree height} = \text{eye height} + \text{distance} \times \tan(\text{angle})$ .
- 04 Plan an efficient route.** Order your checkpoints to minimize backtracking. Note the route on the card.
- 05 Recommend one site.** Pick one sensor location and defend it with your measurements, not opinion.

**Safety:** Buddy system outdoors. Stay on approved route. Sun, hydration, and weather check before launch. Allergy: Pollen sensitivity note. Students do not need to touch plants.

## MATERIALS

|                                          |                                 |
|------------------------------------------|---------------------------------|
| Clipboards                               | 4 (plus 1 spare)                |
| Paper clinometers                        | 4 (plus 2 spares)               |
| Combined temperature and humidity meters | 4 (plus 1 spare)                |
| 3-in-1 soil meters (moisture, light, pH) | 4 (plus 1 spare)                |
| Route cards                              | 4 (one per team, plus 2 spares) |
| Pencils                                  | shared                          |

## YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

**Prediction (what will work best, and why):** \_\_\_\_\_

**Checkpoint readings (temp F / humidity % / light 0 to 2000 / soil 1 to 10):** \_\_\_\_\_

**Estimated tree height (eye height + distance x tan angle):** \_\_\_\_\_

**Our recommended sensor site and the numbers that support it:** \_\_\_\_\_

## DEFEND YOUR RECOMMENDATION

**What measurements made this the best sensor site?** \_\_\_\_\_

**If you ran the route again, what is the ONE thing you would change to get better data?** \_\_\_\_\_

## HOW YOU SCORE (100 POINTS)

| SCORING CRITERION            | POINTS     |
|------------------------------|------------|
| Data accuracy                | 30         |
| Route efficiency             | 20         |
| Site recommendation evidence | 30         |
| Teamwork and field conduct   | 20         |
| <b>Total</b>                 | <b>100</b> |